

Today 9:22 AM

Treasure Hunt

Explore a procedurally generated dungeon, collect the key, then open the chest.

Getting Started

This game uses **pygame**. Install it before running:

```
pip install pygame-ce python game.py
```

Note: If you are on Python 3.12 or newer, use `pygame-ce` (Community Edition) instead of `pygame`. It is a drop-in replacement with the same API.



Controls

Key	Action
W/A/S/D or Arrows	Move
R	Restart

Before changing the code

Play through the dungeon and observe how movement, tile collision, and item collection are handled. Trace the flow from `GameEngine` through `Player` and back. Understand how the grid and player state interact before making any modifications.

Task 1 — Traps

Add floor traps that send the player back to their starting position when stepped on.

Done when: the player is visibly reset to the start room on stepping a trap, and a clear status message is shown.

Task 2 — Enemy Guard

Add a guard that patrols back and forth near the chest. Touching the guard resets the player to start.

Done when: the guard moves between two patrol points each frame and correctly resets the player on collision.

Task 3 — Mini-map

Draw a small mini-map in a corner of the screen showing the dungeon layout.

Done when: the mini-map updates in real time, distinguishes walls from floor, and marks the player's current position.

Task 4 — Inventory UI

Show a HUD inventory slot that is empty at the start and displays a key icon once the key is collected.

Done when: the slot renders correctly in both states and updates immediately on key pickup.

Required testing

Collect the key then the chest, trigger every trap type, collide with the guard, verify the player resets correctly each time, check the mini-map reflects the actual grid, and confirm the inventory slot updates on collection. Test restarting mid-run.

LLM usage

You may use an LLM during the lab. The goal is to use it as a coding assistant while retaining responsibility for understanding and testing the result.

- Inspect the existing code before asking for changes.
- Ask for explanations when you do not understand a proposed change.
- Test generated code against the stated behaviour and edge cases.
- Keep your complete LLM chat history for submission.
- Do not replace the whole project with an unrelated implementation.

Submission checklist

- Tasks 1–4 completed and tested.
- Traps reset the player correctly.

- Guard patrol and collision reset work.
- Mini-map reflects the dungeon in real time.
- Inventory slot updates on key pickup.
- No unnecessary external dependencies added beyond pygame.
- Code remains understandable and modular.
- Complete LLM chat-history link included.

Submission

Submission is only the following three things:

- A 10-second video of gameplay **before** your changes
- A 10-second video of gameplay **after** your changes, showing the new features working
- The Chat/LLM used page link, with the complete chat history so help me out doing this lab

 https://github.com/SETAPESU26/24_treasure_hunt.git

Worked for 9s

no i will use vs code

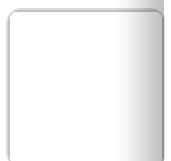


..



Lab_4_VibeCoding...
PDF

I forgot to give this document



done

wt i shud copy paste and where?

```
import pygame
import random

TILE = 40
COLS, ROWS = 20, 15
WALL, FLOOR, CHEST, KEY = 0, 1, 2, 3
SPEED = 3

def generate_world():
    grid = [[WALL]*COLS for _ in range(ROWS)]
    rooms = []
    for _ in range(8):
        w = random.randint(3,6)
        h = random.randint(3,5)
        x = random.randint(1, COLS-w-1)
        y = random.randint(1, ROWS-h-1)
        room = pygame.Rect(x, y, w, h)
        overlap = any(room.inflate(2,2).colliderect(r) for r in rooms)
        if not overlap:
            rooms.append(room)
            for ry in range(y, y+h):
                for rx in range(x, x+w):
                    grid[ry][rx] = FLOOR

    for i in range(len(rooms)-1):
        ax, ay = rooms[i].centerx, rooms[i].centery
        bx, by = rooms[i+1].centerx, rooms[i+1].centery
        cx = ax
        while cx != bx:
            grid[ay][cx] = FLOOR
            cx += 1 if bx > cx else -1
        cy = ay
```

```
while cy != by:
    grid[cy][bx] = FLOOR
    cy += 1 if by > cy else -1

if len(rooms) >= 2:
    cr, ck = rooms[-1], rooms[-2]
    grid[cr.centery][cr.centerx] = CHEST
    grid[ck.centery][ck.centerx] = KEY

start = rooms[0] if rooms else None
return grid, start

COLORS = {
    WALL: (60,50,70),
    FLOOR: (200,190,170),
    CHEST: (200,160,30),
    KEY: (220,220,60),
}

class Player:
    def __init__(self, x, y):
        self.rect = pygame.Rect(x, y, 28, 28)
        self.color = (60,120,220)
        self.has_key = False

    def move(self, keys, grid, rows, cols):
        dx=dy=0
        if keys[pygame.K_LEFT] or keys[pygame.K_a]: dx=-SPEED
        if keys[pygame.K_RIGHT] or keys[pygame.K_d]: dx=SPEED
        if keys[pygame.K_UP] or keys[pygame.K_w]: dy=-SPEED
        if keys[pygame.K_DOWN] or keys[pygame.K_s]: dy=SPEED
        self._try_move(dx,0,grid,rows,cols)
        self._try_move(0,dy,grid,rows,cols)

    def _try_move(self, dx, dy, grid, rows, cols):
        new = self.rect.move(dx,dy)
        for px,py in [(new.left,new.top),(new.right-1,new.top),
```

```
(new.left,new.bottom-1),(new.right-1,new.bottom-1]):
    c,r=px//TILE,py//TILE
    if not(0<=r<rows and 0<=c<cols) or grid[r][c]==WALL:
        return
    self.rect=new
```

```
def draw(self, screen):
    pygame.draw.ellipse(screen, self.color, self.rect)
    if self.has_key:
        pygame.draw.circle(screen, (220,220,60), (self.rect.right-6,
self.rect.top+6), 5)
```

```
WIDTH = COLS * TILE
HEIGHT = ROWS * TILE + 50
FPS = 60
```

```
class GameEngine:
    def __init__(self):
        pygame.init()
        self.screen = pygame.display.set_mode((WIDTH, HEIGHT))
        pygame.display.set_caption("Treasure Hunt")
        self.clock = pygame.time.Clock()
        self.font = pygame.font.SysFont("monospace", 24)
        self.big_font = pygame.font.SysFont("monospace", 40,
bold=True)
        self.reset()
```

```
def reset(self):
    self.grid, start = generate_world()
    if start:
        sx = start.x * TILE + 6
        sy = start.y * TILE + 6
    else:
        sx, sy = TILE+6, TILE+6
    self.player = Player(sx, sy)
    self.won = False
```

```
self.status = "Find the KEY, then the CHEST!"

def handle_events(self):
    for event in pygame.event.get():
        if event.type == pygame.QUIT: return False
        if event.type == pygame.KEYDOWN and event.key ==
pygame.K_r: self.reset()
    return True

def update(self):
    if self.won: return
    keys = pygame.key.get_pressed()
    self.player.move(keys, self.grid, ROWS, COLS)
    pr = self.player.rect.centery // TILE
    pc = self.player.rect.centerx // TILE
    if 0<=pr<ROWS and 0<=pc<COLS:
        cell = self.grid[pr][pc]
        if cell == KEY:
            self.player.has_key = True
            self.grid[pr][pc] = FLOOR
            self.status = "Got the key! Find the CHEST!"
        elif cell == CHEST and self.player.has_key:
            self.won = True
            self.status = "Treasure found!"

def draw(self):
    self.screen.fill((30,25,40))
    for r in range(ROWS):
        for c in range(COLS):
            cell = self.grid[r][c]
            rect = pygame.Rect(c*TILE, r*TILE, TILE, TILE)
            pygame.draw.rect(self.screen, COLORS[cell], rect)
            if cell == KEY:
                pygame.draw.circle(self.screen, (255,240,60),
(c*TILE+TILE//2, r*TILE+TILE//2),10)
            elif cell == CHEST:
                pygame.draw.rect(self.screen,
```

```

(180,120,20),rect.inflate(-12,-12),border_radius=4)
    self.player.draw(self.screen)
    hud = pygame.Rect(0,ROWS*TILE,WIDTH,50)
    pygame.draw.rect(self.screen,(20,20,35),hud)
    st = self.font.render(self.status+" | R=Restart", True,
(200,200,200))
    self.screen.blit(st,(8,ROWS*TILE+13))
    if self.won:
        ov=pygame.Surface((WIDTH,ROWS*TILE),pygame.SRCALPHA)
        ov.fill((0,0,0,140))
        self.screen.blit(ov,(0,0))
        msg=self.big_font.render("TREASURE FOUND!", True,
(220,180,30))
        sub=self.font.render("Press R to Play Again",True,
(180,180,180))
        self.screen.blit(msg,(WIDTH//2-
msg.get_width()//2,ROWS*TILE//2-30))
        self.screen.blit(sub,(WIDTH//2-
sub.get_width()//2,ROWS*TILE//2+20))
        pygame.display.flip()

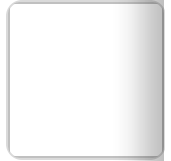
    def run(self):
        running=True
        while running:
            running=self.handle_events()
            self.update()
            self.draw()
            self.clock.tick(FPS)
        pygame.quit()

if __name__ == "__main__":
    engine = GameEngine()
    engine.run()

```

so before changing anything in game.py wt they expect from us is ,
 fork into my github account ,clone the forked repo, video,make
 changes and commit and push to my repo,video after the changes

see i already have se lab repo shud i fork it insdide of it mentioning my file name as lab4_24_treasure_hunt or seperate fork i do ?



done



kk video is saved

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git status
On branch main
Your branch is up to date with 'upstream/main'.
```

nothing to commit, working tree clean

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git log -1 --oneline
506ca0d (HEAD -> main, upstream/main, upstream/HEAD) Initial
commit
```

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt>
```

done

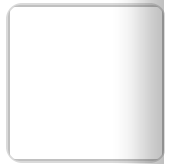
done

is it is correct



done

done



?

done

no it did not crashh

Treasure Hunt Task 1 testing result: I can see red traps, but have not yet confirmed collision reset. Guide me through the next debugging or verification step.

no when i catch the crash it will open a game which has no trap at all

Today 9:05 PM

like wt shud i change and exactly where?

```
PS E:\poppi\sem5\SE\LL> python game.py
C:\Users\deepg\AppData\Local\Programs\Python\Python310\python
.exe: can't open file 'E:\\poppi\\sem5\\SE\\LL\\game.py': [Errno 2] No
such file or directory
PS E:\poppi\sem5\SE\LL>
```



RECORD_2.mp4
File

IS THIS ONE IS KK?

Worked for 23s

after change video

no he return to starting point now

no i g

sry my bad yeah it comes

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git status
```

On branch main

Your branch is up to date with 'upstream/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: game.py

no changes added to commit (use "git add" and/or "git commit -a")

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> python game.py
```

pygame-ce 2.5.8 (SDL 2.32.10, Python 3.10.0)

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> python game.py
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pygame-ce 2.5.8 (SDL 2.32.10, Python 3.10.0)

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git status
```

On branch main

Your branch is up to date with 'upstream/main'.

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: game.py

no changes added to commit (use "git add" and/or "git commit -a")

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git add game.py
```

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git commit -m "Task 1:"
```

Add floor traps and reset behavior"

[main e45b2c8] Task 1: Add floor traps and reset behavior

1 file changed, 45 insertions(+), 13 deletions(-)

PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git push -u origin main

Enumerating objects: 5, done.

Counting objects: 100% (5/5), done.

Delta compression using up to 12 threads

Compressing objects: 100% (3/3), done.

Writing objects: 100% (3/3), 813 bytes | 813.00 KiB/s, done.

Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (2/2), completed with 2 local objects.

To https://github.com/GTX-Pradeep/24_treasure_hunt.git

506ca0d..e45b2c8 main -> main

branch 'main' set up to track 'origin/main'.

PS E:\poppi\sem5\SE\LL\24_treasure_hunt> SEE IS IT'S CORRECT?

YEAH I LOCATED

DONE

DONE

```
def __init__(self):
    pygame.init()
    self.screen = pygame.display.set_mode((WIDTH, HEIGHT))
    pygame.display.set_caption("Treasure Hunt")
    self.clock = pygame.time.Clock()
    self.font = pygame.font.SysFont("monospace", 24)
    self.big_font = pygame.font.SysFont("monospace", 40,
bold=True)
    self.reset()

def reset(self):
    self.grid, start = generate_world()
    if start:
        sx = start.x * TILE + 6
```

```
sy = start.y * TILE + 6
else:
    sx, sy = TILE+6, TILE+6
self.player = Player(sx, sy) HERE ONLY RIGHT?
```

DONE

DONE

done

done working

do i have to make a video of it?

done

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git status
On branch main
Your branch is up to date with 'origin/main'.
```

nothing to commit, working tree clean

```
PS E:\poppi\sem5\SE\LL\24_treasure_hunt> git log -2 --oneline
e0a0951 (HEAD -> main, origin/main) Task 2: Add patrolling enemy
guard
e45b2c8 Task 1: Add floor traps and reset behavior
PS E:\poppi\sem5\SE\LL\24_treasure_hunt>
```

so see will finish of quickly

done